

# Towhid Ahmed

AI Engineer · Open to remote — worldwide

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## Professional Summary

Junior AI Engineer with 1+ year of freelance experience building AI and Generative AI applications using Python, Machine Learning, FastAPI, LLMs, RAG, LangChain, and LangGraph. Built 3 production-oriented AI applications, improving retrieval relevance by 25%, reducing hallucinations by 35%, and achieving 98/98 passing automated tests. Experienced in RAG pipelines, LLMops, Docker, and LangSmith.

## Technical Skills

**Programming & Software Engineering:** Python, SQL, OOP, Async Programming, Data Structures & Algorithms

**Machine Learning:** Supervised & Unsupervised Learning, Scikit-learn, Feature Engineering, Model Evaluation

**Deep Learning:** Neural Networks, TensorFlow

**Generative AI:** LLMs, RAG, AI Agents, Prompt Engineering, LangChain, LangGraph

**NLP & Transformers:** NLP, Transformers, Hugging Face, Sentence Transformers, Embeddings

**LLMs & AI Platforms:** OpenAI, Llama, Groq

**Databases & Backend:** Pinecone, PostgreSQL, FastAPI, REST APIs

**Deployment & Observability:** Docker, Git, CI/CD, Pytest, LangSmith, LLM Guardrails

## Professional Experience

**Freelance AI Engineer | Self-Employed | Jun 2025 – Present**

**AI Document Assistant | Jun 2026 – Jul 2026**

- Enhanced information reliability and user trust by developing a **3-agent LangGraph workflow** with **citation-based RAG**, significantly **reducing AI hallucinations**.
- Powered seamless document processing and interaction by engineering **7 production-ready REST APIs** using **FastAPI, OpenAI API, and Pinecone**.
- Enabled **1,000+** document knowledge-base updates without redeployment.

**AI Sales & Customer Support Assistant | Aug 2025 – Sep 2025**

- **Automated customer interactions and improved consistency** using a multi-agent **LangGraph pipeline** (Intent, Retrieval, Response) with Pinecone RAG and **conversational memory**.
- Slashed AI inference costs and maintained high execution speed by integrating **Groq**, Hugging Face embeddings, and **Pinecone Serverless**.
- **Built a scalable support infrastructure** with a Streamlit UI and a FastAPI backend featuring **6+ REST APIs** for knowledge uploads and session management.

## Selected Projects

**Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, Hugging Face, Tavily** GitHub | Demo

- **Designed 4-agent LangGraph workflow** routing queries between **RAG** and **web search** based on intent and complexity.
- Built **FastAPI** backend with async execution, **7 core APIs** (/chat, /upload, /health, /metrics) optimized for **Render**.
- Integrated **Tavily with DuckDuckGo fallback** and **LangSmith tracing** for reliable web retrieval and **observability**.

**Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching** GitHub

- Designed production multimodal retrieval (PDF/audio/video) **improving relevance by 25%** and **reducing hallucinations by 35%**.
- Implemented two-stage retrieval (**semantic chunking** + cross-encoder **re-ranking**) with **<2s latency** via **caching**.
- Tech: **Groq Llama 3.3, BAAI embeddings, Pinecone, FastAPI, Streamlit, Docker**; 76-unit test coverage.

**Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT** GitHub

- Built a clinical information assistant using a **4-agent LangGraph pipeline** (Router, Safety, Retrieval, Answer) with **RAG and web search**.
- Implemented **multi-layer safety system** (input filters, retrieval validation, output sanitization) with JWT auth.
- Achieved **98/98** passing tests across **guardrails, RAG evaluation, and API workflows**.

## Education and Certificates

**Education: Computer Science Engineering(CSE)**, Chittagong University of Engineering & Technology | 2023-2027

**Certifications:** • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMops — Udemy • Data Structures & Algorithms — TakeUForward